UNIVERSITY OF CAMBRIDGE INTERNATIONAL EXAMINATIONS

GCE Ordinary Level

MARK SCHEME for the October/November 2008 question paper

5054 PHYSICS

5054/04 Paper 4 (Alternative to Practical), maximum raw mark 30

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1 **(a)** 21 cao B1 [1]

(b) any one of following:

allow answer correctly drawn on Fig 1.1

scale/ruler placed close to tube

eye (level)/sight perpendicular to ruler/scale/reading **NOT** meniscus

eye level with reading/meniscus B1 [1]

(c) axes: correct way round, labelled quantity and unit B1

scales: more than ½ grid, sensible, at least three values B1

points: plotted accurately, to ½ smallest square, maximum size 1 mm dots B1

line: best fit, straight neatly drawn with ruler; allow joints first to last points B1 [4]

(d) pressure not zero when not in liquid/when $d = 0$ /

p = 3 to 6 cm water when $d = 0$ /

zero error in reading/measuring pressure /

pressure changed when apparatus assembled B1 [1]

(e) d increases (equal amounts) as p increases (by equal amounts) /

linear relationship / straight line not through 0,0/positive gradient B1 [1]

NOT directly proportional / doubling d doubles p / linearly proportional / p proportional to d

(f) 18.5 to 19.5 ecf graph B1 [1]

NOT if no line through points drawn on graph

[Total: 9]

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2	**(a)**	line through OP_1P_2 with ruler, neat, touching each dot	B1	
		line through P_3P_4 with ruler, neat, touching each dot	B1	
		lines produced to meet	B1	[3]
	(b)	repeat for Q	B1	[1]
	(c)	mirror line accurately drawn through intersections	B1	[1]
	(d)	P_3P_4 and Q_3Q_4 produced back to meet	B1	[1]
	(e)	12.5 to 14.0 cm to nearest mm unit required	B1	[1]
	(f)	pins may not be vertical / pins not straight / where pins enter paper		
		allow more accurate / pins placed correctly or properly	B1	[1]
				[Total: 8]
3	**(a)**	reading on scale not zero when closed / no object	B1	[1]
	(b)	28 seen or implied	C1	
		1.28 seen / 8 seen	C1	
		0.16 cao unit not required (0.427 scores 2)	A1	[3]
	(c)	easier to use (e.g. no vernier to read/automatic reading) / reduces operator error / quicker to use / more sf or dp or precise	B1	[1]
		uses a battery / cell / battery / cell may run out / expensive / too sensitive / readings fluctuate / (circuit) malfunction	B1	[1]
				[Total: 6]

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4	**(a)**	**(i)**	dim(mer than normal) / not lit	B1	[1]
			normal brightness / brighter (than A) / bright	B1	[1]
		(ii)	(glows brightly then) blows / filament fuses / melts	M1	
			voltage / current (too) high / more than working value/6V	A1	[2]
	(b)		variable resistor / variable power supply / changing number / voltage of cells drawn or described	B1	
			voltmeter across lamp / power supply drawn or described	B1	
			start with low V/high R then measure V just before lamp blows	B1	[3]
			max 2 if no circuit drawn showing lamp and power supply / incorrectly drawn symbols / additional incorrectly connected components		

[Total: 7]

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